

PRD Quality Reviewer

Sample PRD Document

Real-Time Fraud Detection for Payment Gateway

Problem Statement

Our payment gateway processed \$2.4B in volume last year with a 0.7% fraud rate (industry avg: 0.3%). We lost an estimated \$16.8M to fraud. Current batch-processing detection (24-hour delay) is insufficient.

Business Objectives

- 1. Reduce fraud rate from 0.7% to <0.35% within 6 months of launch
- 2. Maintain false positive rate below 2%
- 3. Keep P95 latency under 200ms for payment authorization

Success Metrics

Metric	Current	Target
Fraud rate (% of volume)	0.7%	<0.35%
False positive rate	~3%	<2%
Auth latency (P95)	120ms	<200ms
Engineering cost	N/A	<\$25K/month (GCP)

Target Users

- Primary: Payments risk team (5 analysts)
- Secondary: End customers (transparent experience)

Feature Requirements

F1: Real-Time Scoring Engine

- Score each transaction (0-100) at authorization time
- Features: velocity, device fingerprint, geolocation, amount deviation
- Rules engine override for known-good merchants

F2: Decision Orchestration

- Score < 30 !' auto-approve
- Score 30-70 !' 3DS challenge (step-up auth)
- Score > 70 !' decline with reason code
- Analyst override within 5-minute window for VIP customers

F3: Feedback Loop

- Analysts mark false positives/negatives in dashboard
- Labeled data retrains model weekly
- A/B test framework: model v2 vs v1 on 5% shadow traffic

Non-Requirements

- Chargeback representment (existing system)
- PCI-compliant storage (separate initiative)

Timeline

Milestone	Date	Deliverable
Real-time scoring API	Week 6	POC, manually tested
Decision orchestration	Week 8	Full flow
ML model v1	Week 10	Trained on 12 months data
Shadow mode	Weeks 11-12	5% traffic
Phased rollout	Weeks 13-15	10% !' 25% !' 50% !' 100%

Risks

Risk	Likelihood	Mitigation
Latency spike from ML	Medium	Fallback to rules-only via feature flag
Model drift	High	Weekly retraining + drift detection

| High false positive rate | Medium | Analyst override + manual review queue